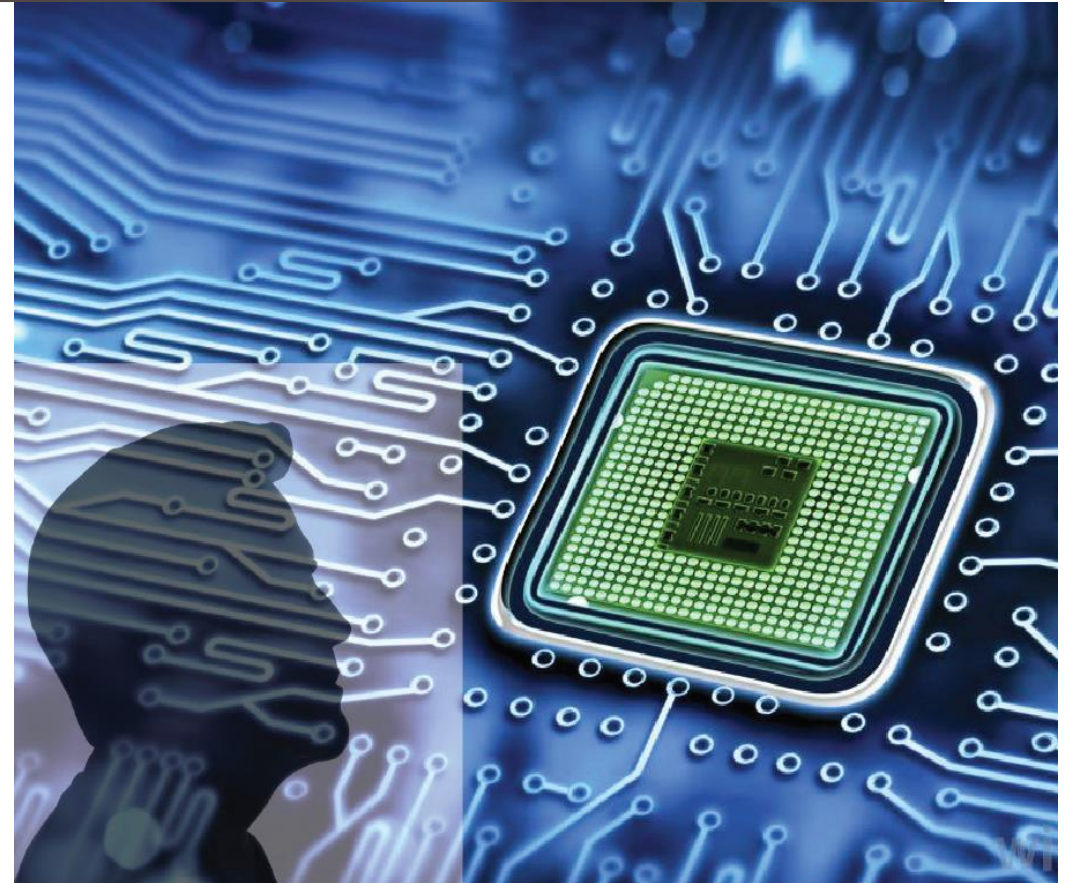


# VLSI DESIGN FLOW

## Week-14 Lecture-1 Physical Design

Sneh Saurabh  
25<sup>th</sup> November, 2024



# VDF: Revision/Announcements

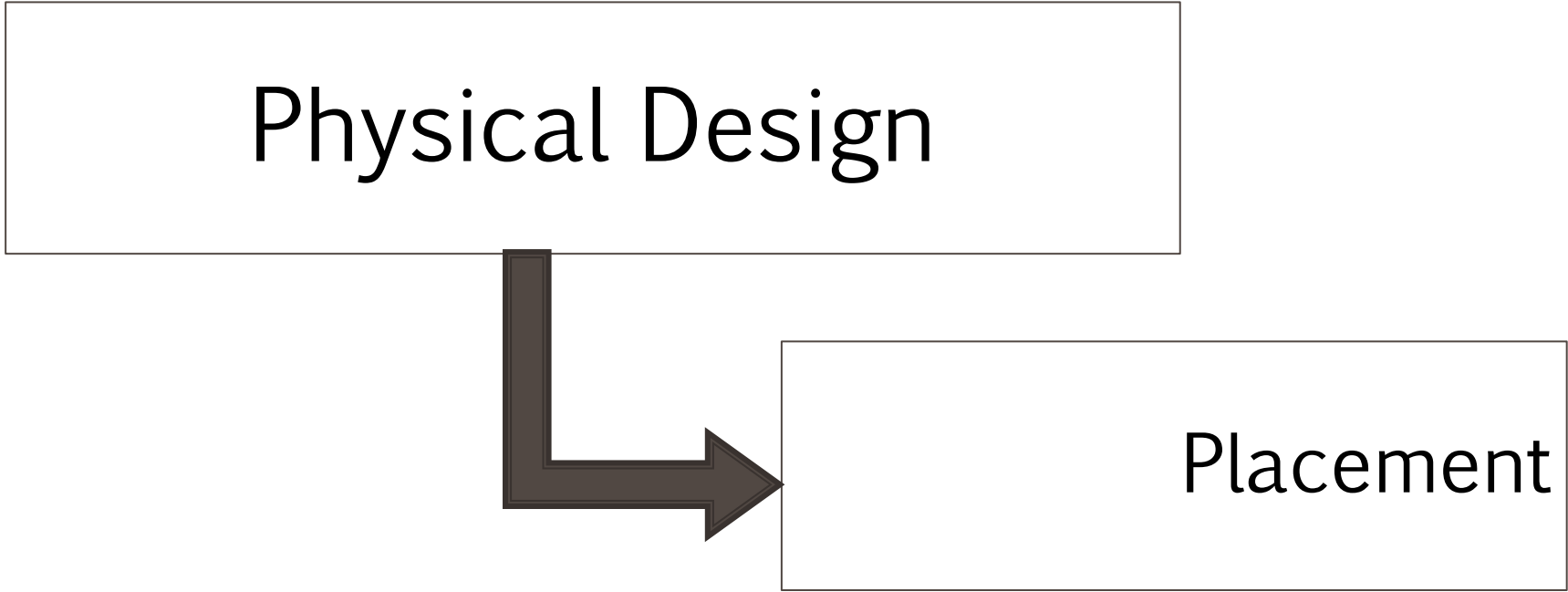
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graph TD; A[Physical Design] --> B[Placement]
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Physical Design

Placement

# Spare Cell Placement

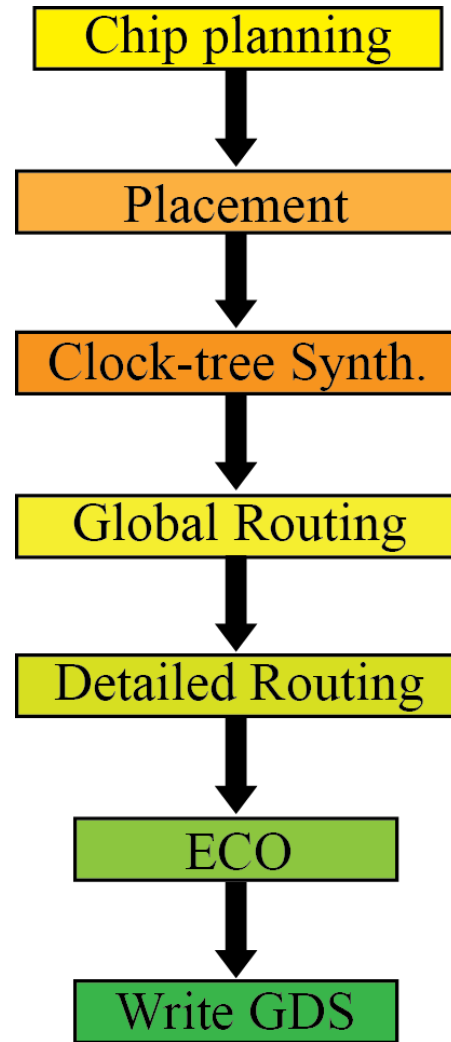
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## Spare cells:

- Extra cells put in the circuit in anticipation of later use
  - After fabrication, if it is required that some cells need to be connected, then spare cells can be used
    - Only the top metal layers might need to be changed
- 
- Cannot be anticipated where will be the spare cells be actually be required to be connected
    - Place randomly over the unused placement area
  - Tools must be informed that which are spare cells and should not be optimized out

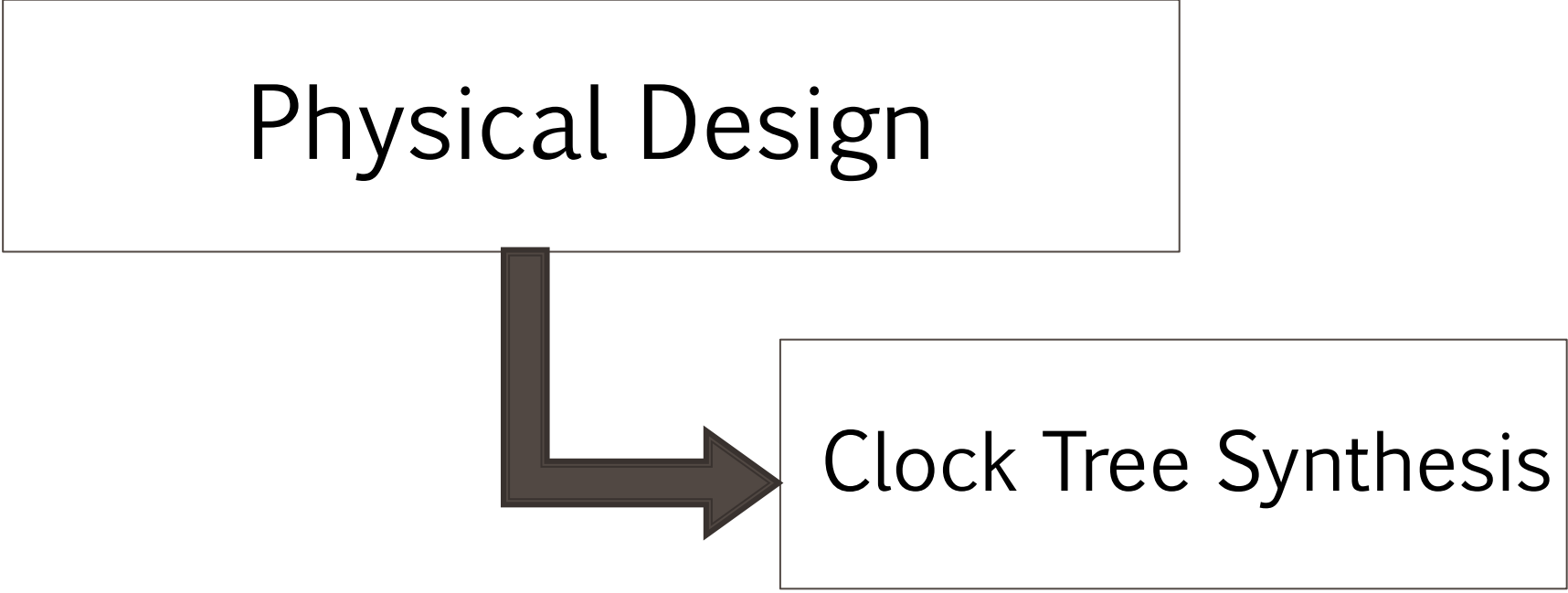
# Physical Design

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graph TD; A[Physical Design] --> B[Clock Tree Synthesis]
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Physical Design

Clock Tree Synthesis

# Clock Tree Synthesis (CTS): Basics

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## Ideal clock signal:

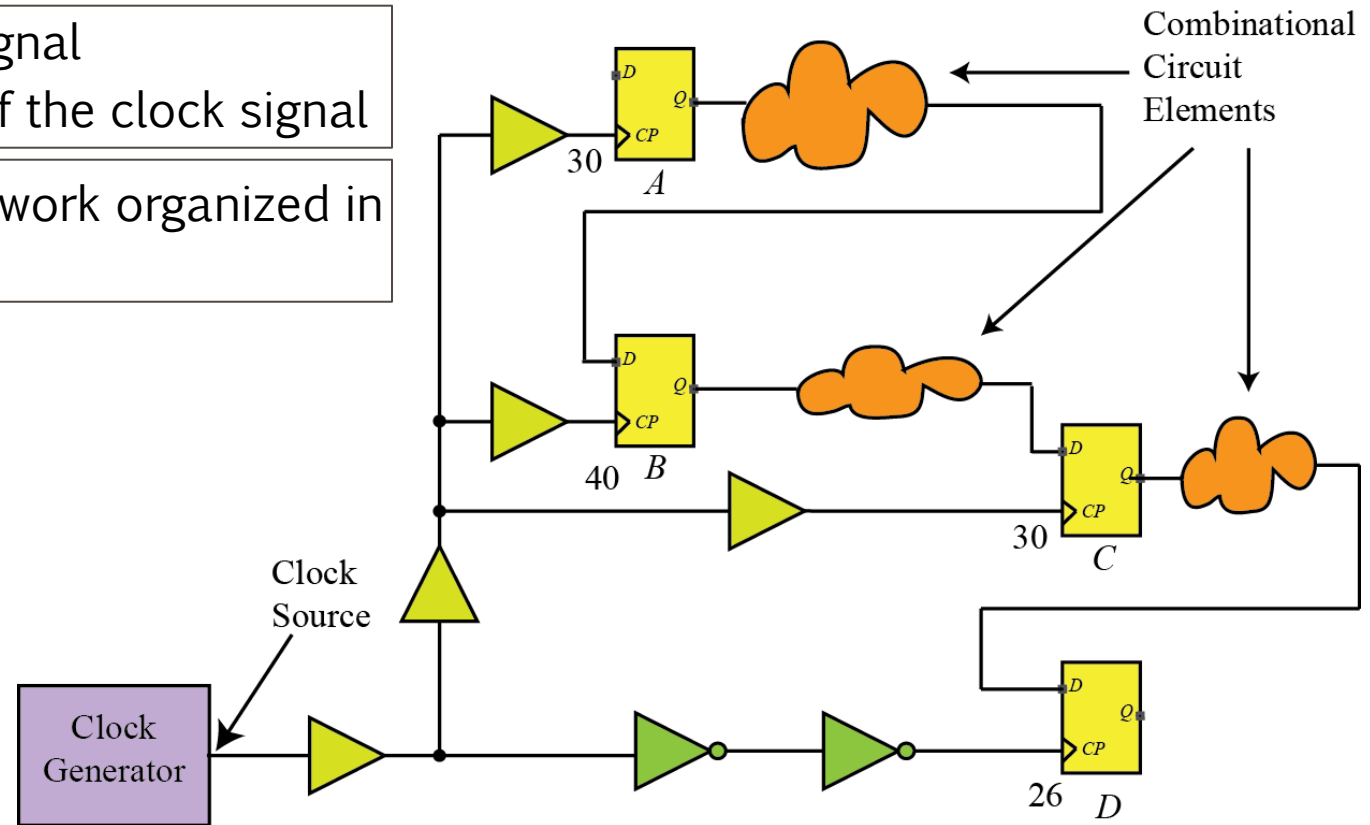
- Has the same waveform (voltage vs. time relationship) at all the points in the circuit.
  - No clock skew or no difference in the arrival time of the clock signal at the flip-flops
- Clock Tree Synthesis implements clock distribution structure on the layout having behavior similar to the ideal clock
  - Minimizes the clock skew
  - Inserts clock buffers/inverters and performs routing of clock distribution network
- Targets power reduction
  - Clock network consumes significant portion of total active power (25% – 70%)
- Fixes timing violations and signal integrity issues

# CTS: Terminologies (1)

**Source:** starting point of clock signal

**Sinks:** final receiving endpoints of the clock signal

**Clock Tree:** clock distribution network organized in the form of a tree



**Insertion Delay/delay/latency:** time taken by the clock signal to propagate through a clock tree and reach a clock sink



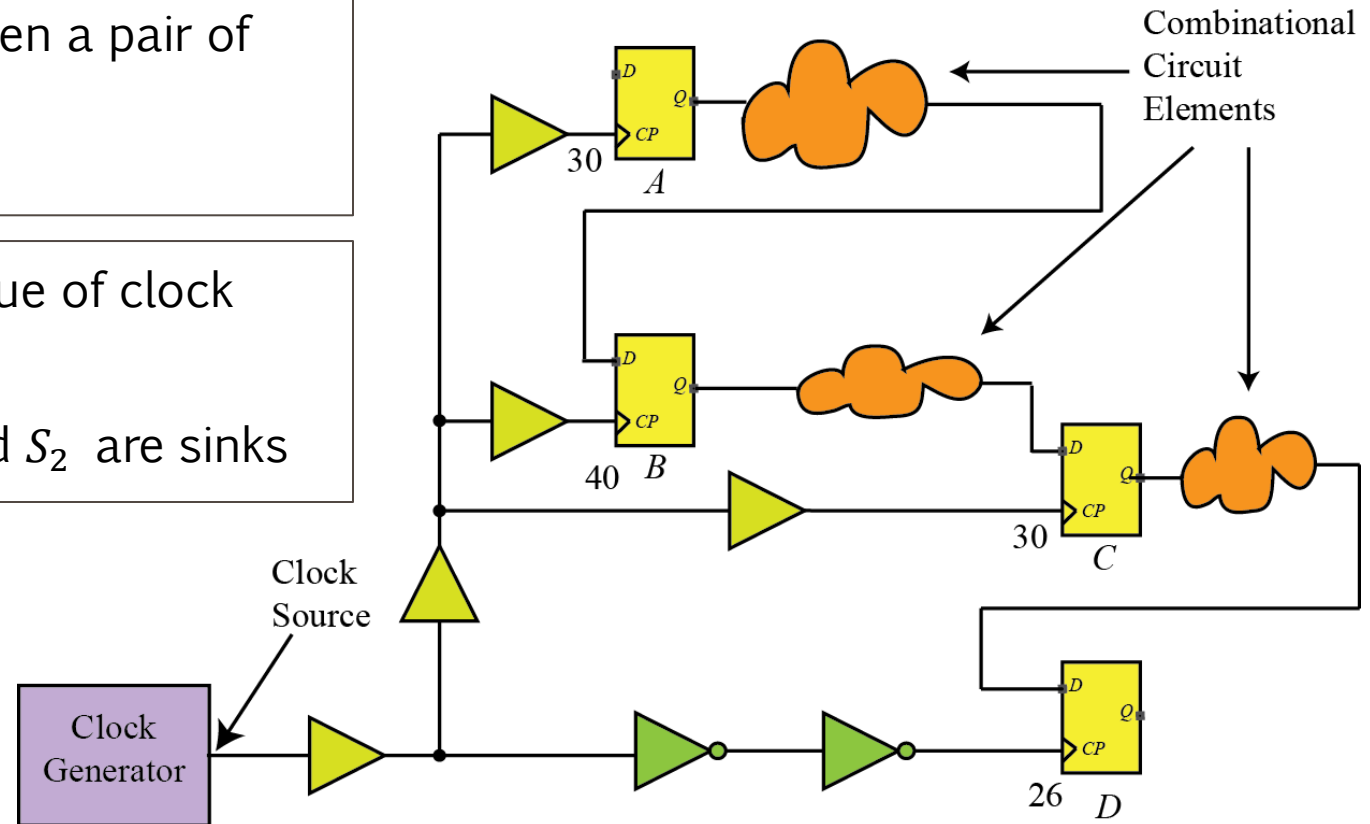
## CTS: Terminologies (2)

**Clock Skew between two sinks:** difference in the arrival time of clock signal between a pair of sinks  $S_1$  and  $S_2$ , i.e.,

$$\delta_{S_1 S_2} = t_{S_1} - t_{S_2}$$

**Global Clock Skew:** maximum value of clock skew between any pair of sinks

$$\Delta_{global} = \text{Max}|\delta_{S_1 S_2}| \text{ where } S_1 \text{ and } S_2 \text{ are sinks}$$



# Clock Distribution Network

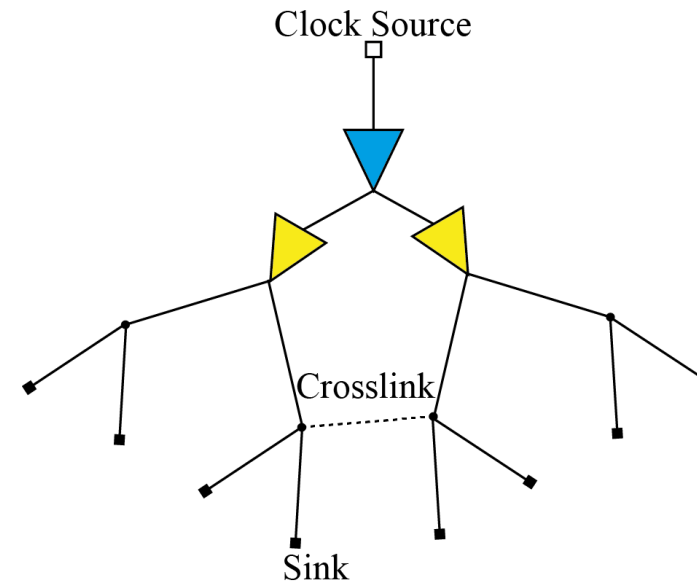
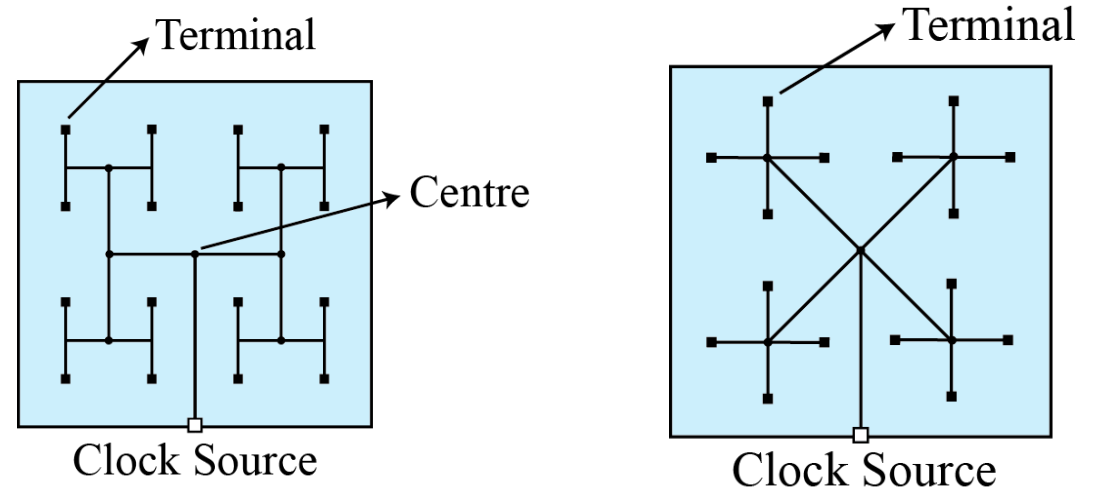
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# Symmetric Tree Architecture

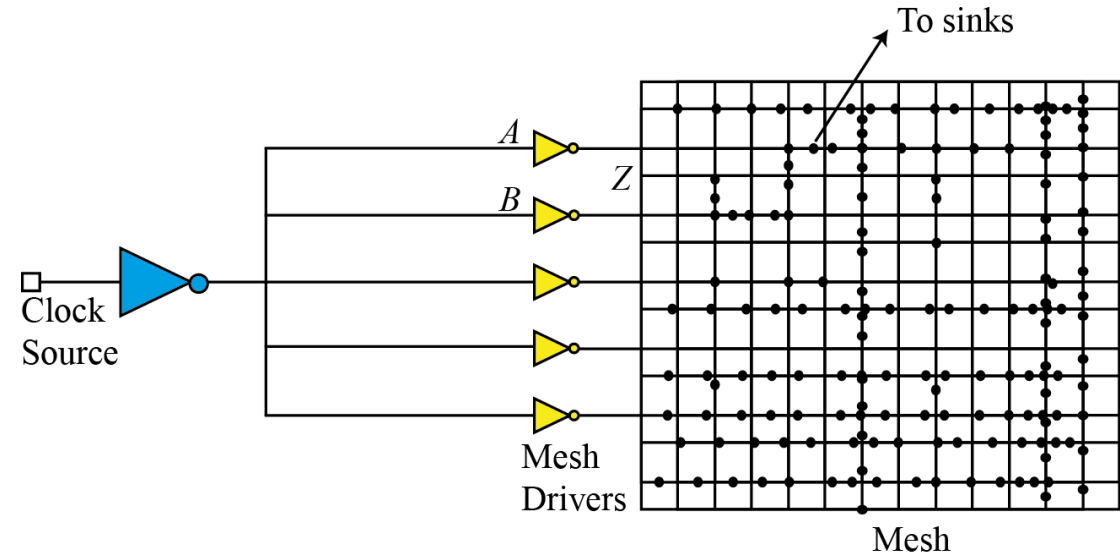
- Clock is first routed to a central point in the chip
- From central point another symmetric architecture forks out
- Example: H-Tree, X-Tree

- Ideally if each path is balanced, skew will be zero
- Due to PVT variations, skew is still observed
- Non-tree architecture can reduce impact of PVT variations



# Mesh Architecture

- Create 2-D structure with redundant wires and devices
- Ensures more paths between mesh drivers and clock sinks
  - Very small skew
  - Decreased impact of process induced variations
  - Robust



## Disadvantages:

- Increases total capacitance and the size of clock drivers
- Increases power consumption (due to increased capacitance and short-circuit power dissipation)

# Useful Skews

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- Introducing well-controlled skews can improve the system performance (useful skews)

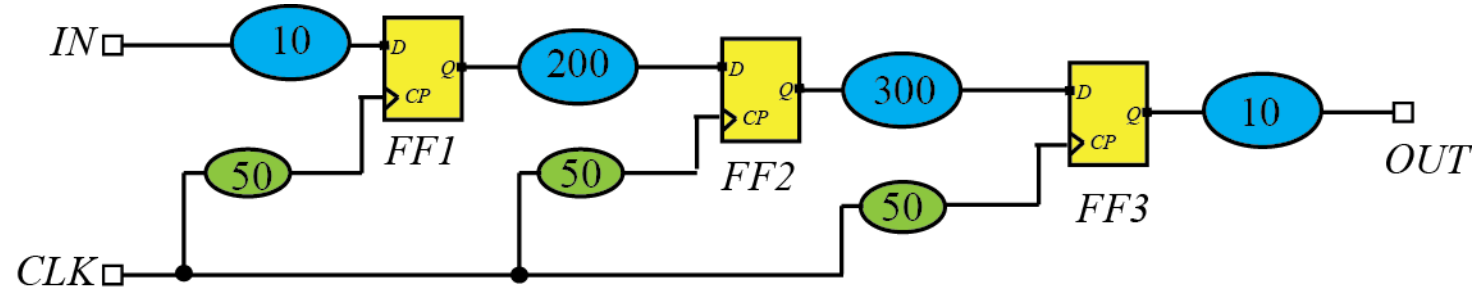
## When can clock skews be useful?

- If a significant difference in the slack exists on the two sides of a flip-flop

## Technique:

- Allocate the excess margin on one side of the flip-flop to the more critical path side
  - Provide more time for data to propagate on the critical path

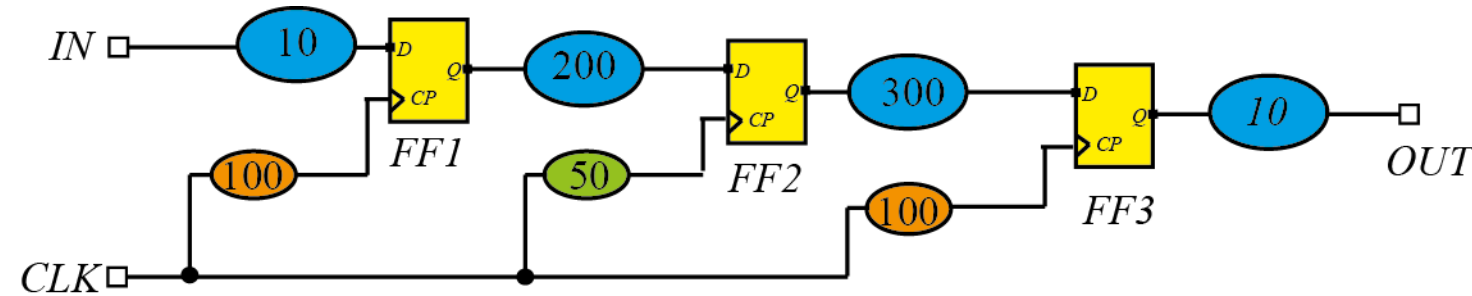
# Useful Skews: Illustration



$$50 + 200 < T_{period} + 50 \\ \Rightarrow 200 < T_{period}$$

$$50 + 300 < T_{period} + 50 \\ \Rightarrow 300 < T_{period}$$

$$f_{max} = \frac{1}{300ps} = 3.33 \text{ GHz}$$



$$100 + 200 < T_{period} + 50 \\ \Rightarrow 250 < T_{period}$$

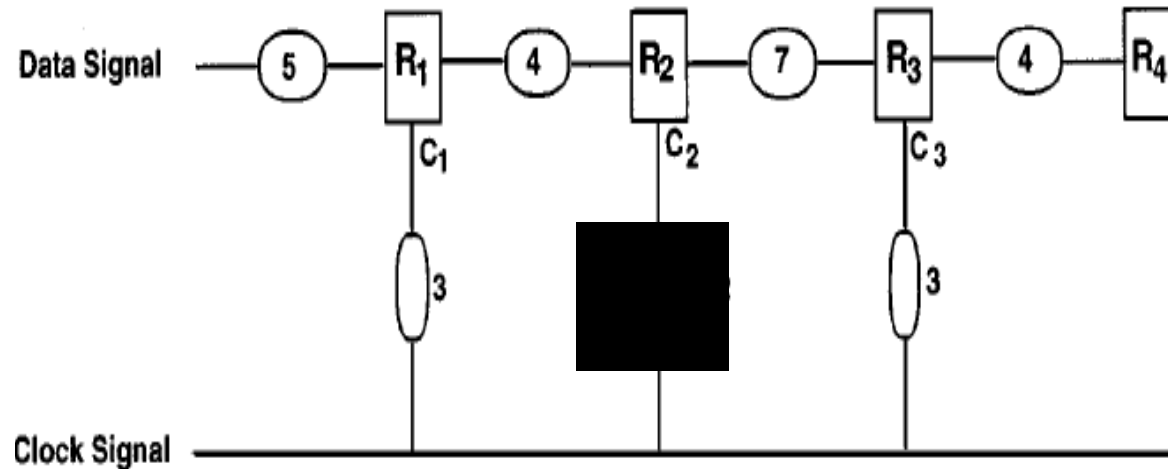
$$50 + 300 < T_{period} + 100 \\ \Rightarrow 250 < T_{period}$$

$$f_{max} = \frac{1}{250ps} = 4 \text{ GHz}$$

# Useful Skew: Example

## Problem:

Compute the delay of the element shown by black-box to make the setup slack at R2 equal to the setup slack at R3. Assume setup time is 0.



Source: Friedman, Eby G. "Clock distribution networks in synchronous digital integrated circuits." *Proceedings of the IEEE* 89.5 (2001): 665-692.

# Obtaining Useful Skews

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- CTS tools calculates useful skew targets for all critical registers
  - Increase arrival time of the clock for target registers by addition of delay buffers or extra wires
- 
- Useful skew also helps in reducing voltage drop problems
    - Registers do not switch at the same time
    - Simultaneous current drawn is less



# CTS: Post Optimization

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- Before CTS clocks were ideal
- After CTS clock network is fully built
- Need to inform STA tool to use actual delays on the clock path

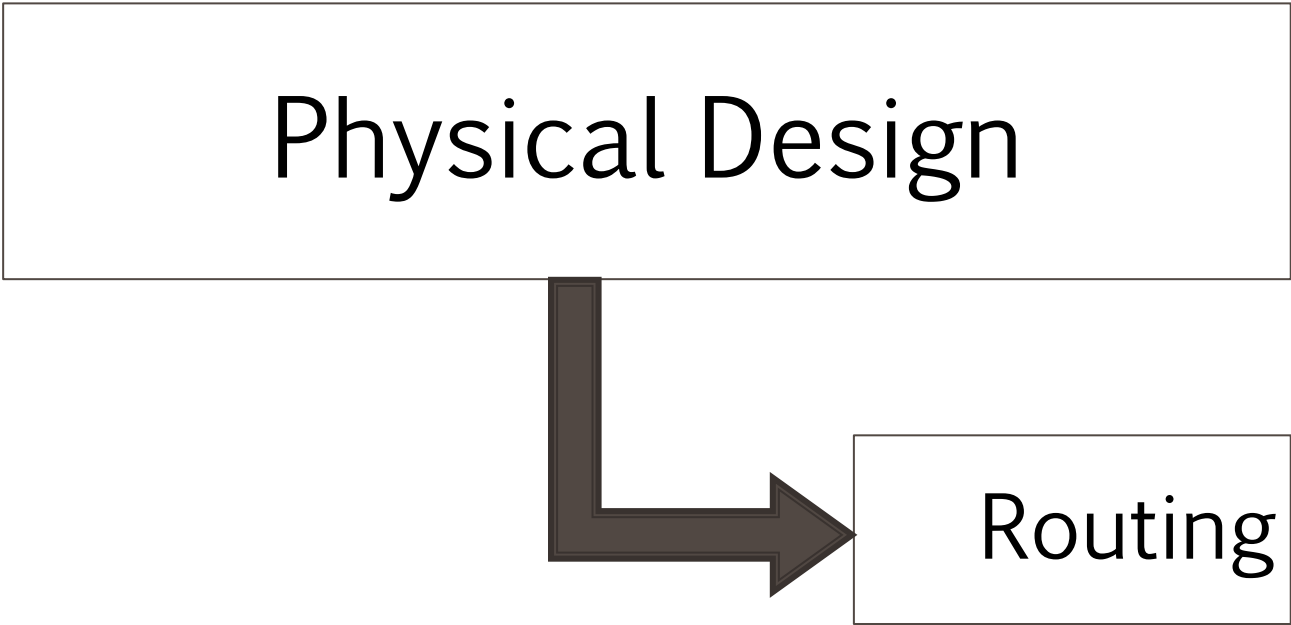
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set_propagated_clock [all_clocks]
```

- New timing violations may show up after CTS
  - Due to additional skew
  - Due to some logic in data path moved out (placement changed) due to addition of buffers in clock network

- Clock Network is frozen after post-CTS optimization and very small ECO fixes are allowed

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graph TD; A[Physical Design] --> B[Routing]
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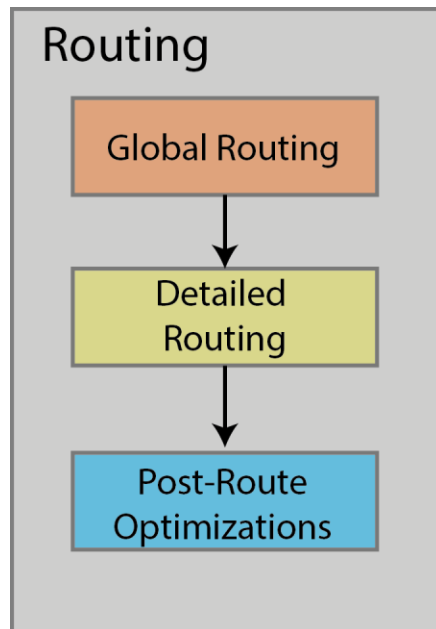
Physical Design

Routing

# Routing: Basics

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- Involves making physical interconnections between different components of a design
- Honor connectivity as in the given netlist
- Complicated and time-consuming task.
- Tight constraints (routing resources, design rules, timing, and signal integrity constraints)



## Global Routing

- Creates the plan of routing for each net (in terms of *routing regions*)
- Actual layout of nets not created

## Detailed Routing

- Decides actual layout of each net in the pre-assigned *routing regions*

## Post-Routing Optimizations

- Localized changes to fix issues in the design

# Global Routing: Goals and Objectives

## Goal:

Provide complete instructions to the detailed router on where to route each net in a design

## Objectives

- Maximize the probability that the detailed router can complete the routing
- Minimize Total Interconnect Length
- Minimize critical path delay

## Desirable:

- Global Routing used in other design phases for estimates: prototyping, floorplanning, placement etc.
  - Need to be fast

# Global Routing: Routing Model

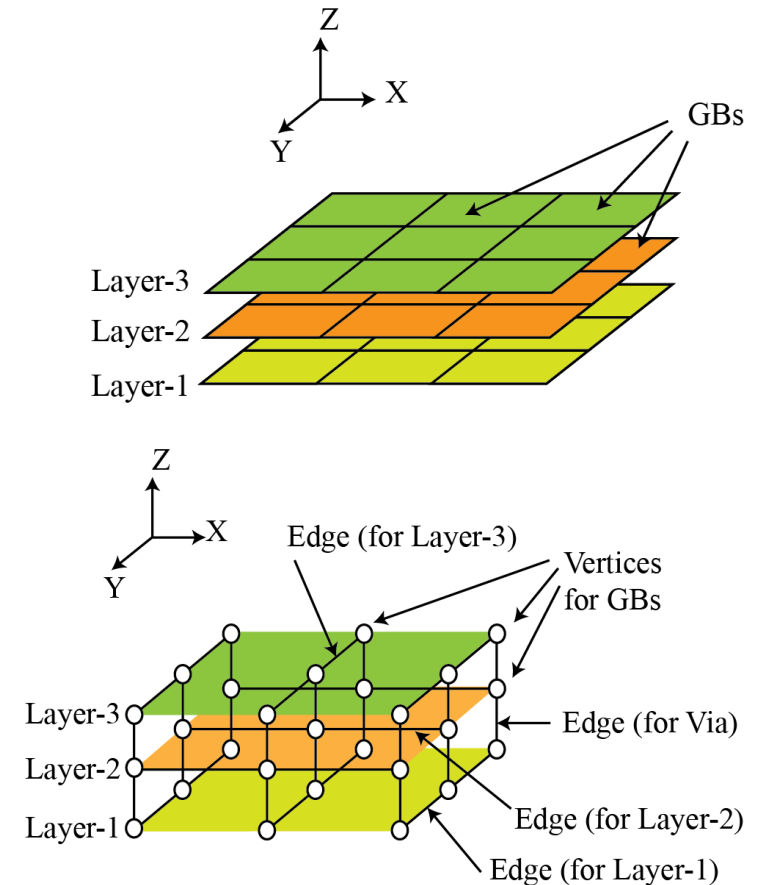
- Routing area is divided into rectangular grids
- Each rectangular region is called **Global Bin** (GB or global tile, routing tile global cell, or bucket)

**Grid graph** is built using GBs:

- Vertices ( $v$ ): GBs
- Edges ( $e$ ): represent boundary between adjacent GBs
  - Create edges between two adjacent vertices only if they lie along the preferred direction of routing
  - Create edges between two vertices that lie vertically adjacent (for vias)

**Determining a route for a net:**

- Associate a pin with a vertex in the grid graph using the GB in which the pin is placed
- Find an *optimal* path in the grid graph that contains all the vertices for that net



# References

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- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.